

# Manmohan Mandhan

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## Education

### National Institute of Technology, Rourkela

Dec 2021 – Jun 2025

B.Tech. in Computer Science and Engineering | CGPA: 7.77/10.0

*Relevant Coursework:* Data Structures and Algorithms, DBMS, OS, Computer Networks, Software Engineering

## Professional Experience

### Software Engineer I, American Express – Bangalore, India

Aug 2025 – Present

- Led development of **Vega**, an in-house automation platform replacing AppViewX (saving \$3M+ annually in licensing costs); built full-stack solution using **Java microservices, Python (Django), and React** processing 500+ Disaster Recovery Exercises monthly.
- Designed and developed **analytics dashboards with real-time data visualization** tracking SSL certificate lifecycle, expiration status, and revocation metrics; integrated backend APIs and cron jobs to maintain fresh data for 20000+ certificates.

### Software Dev Engineer Intern, Amazon Web Services (AWS) – Hyderabad, India

Jan 2025 – Jun 2025

#### [Certificate](#)

- Automated multi-region infrastructure deployment for IoT SiteWise Business Insights using **AWS CDK and CloudFormation**, reducing deployment time from 4 hours to 30 minutes across 16 AWS regions with CI/CD pipeline.
- Designed **AWS Glue Streaming ETL pipeline** with end-to-end encryption (KMS, Secrets Manager) processing 10M+ IoT events daily into Amazon Redshift; implemented optimized UPSERT operations and authored HLD/LLD documentation.
- Implemented CloudWatch monitoring and alerting with comprehensive test suite (pytest), reducing mean time to detection from 45 to 10 minutes and created technical onboarding documentation.

### Software Engineer Intern, American Express – Gurugram, India

May 2024 – Jul 2024

#### [Certificate](#)

- Developed full-stack **Severity Ticketing Portal using React and Node.js**, streamlining incident management workflows for 500+ users across 3 vendor teams (Expereo, Airtel, Tata Communications) and reducing average ticket creation time from 5 to 2 minutes.
- Designed optimized API architecture analyzing vendor API documentation to minimize server load and call frequency; built **Django database synchronization layer** ensuring data consistency between vendor systems and internal DB while maintaining high availability.

## Technical Projects

### Project Phoenix – Competitive Programmer's Platform [GitHub](#)

Apr 2023 – Jul 2023

*Next.js, React, Python, Firebase, REST APIs*

- Built full-stack platform aggregating real-time contest data from multiple APIs (Codeforces, LeetCode) serving 200+ active users; implemented Firebase authentication, real-time notifications, and centralized resource management with curated learning materials.

## Technical Skills

**Programming:** Java, Python, JavaScript, TypeScript, C++, SQL

**Backend & Infrastructure:** Java Microservices, Django, Node.js, Express.js, REST APIs, Event-driven Architecture

**Cloud & DevOps:** AWS (CDK, CloudFormation, Lambda, Glue, Kinesis, Redshift, CloudWatch, KMS, S3), Infrastructure as Code, Docker, Git, CI/CD

**Databases:** PostgreSQL, MongoDB, Amazon Redshift, Firebase (SQL query optimization, index design, NoSQL data modeling)

**Distributed Systems:** Fault tolerance, High availability, Horizontal scalability, Multi-region deployment, Asynchronous processing

**Problem Solving:** 1400+ problems solved | LeetCode (350+), Codeforces (650+), CodeChef (400+)

## Achievements & Leadership

- **Vice President**, Algorithmic & Programming Society, NIT Rourkela (2023-24): Led technical society growing participation from 50 to 300+ students; founded AlgoUtsav 2024 with 1000+ participants from 55+ colleges and secured 4 sponsors.
- **IEM Hacks 2.0:** Winner – Ed-Tech Theme among 200+ teams (2023) [Certificate](#)
- **Google Kickstart:** Global Rank 2232 in Round H among 110,000+ participants (2022) [Certificate](#)